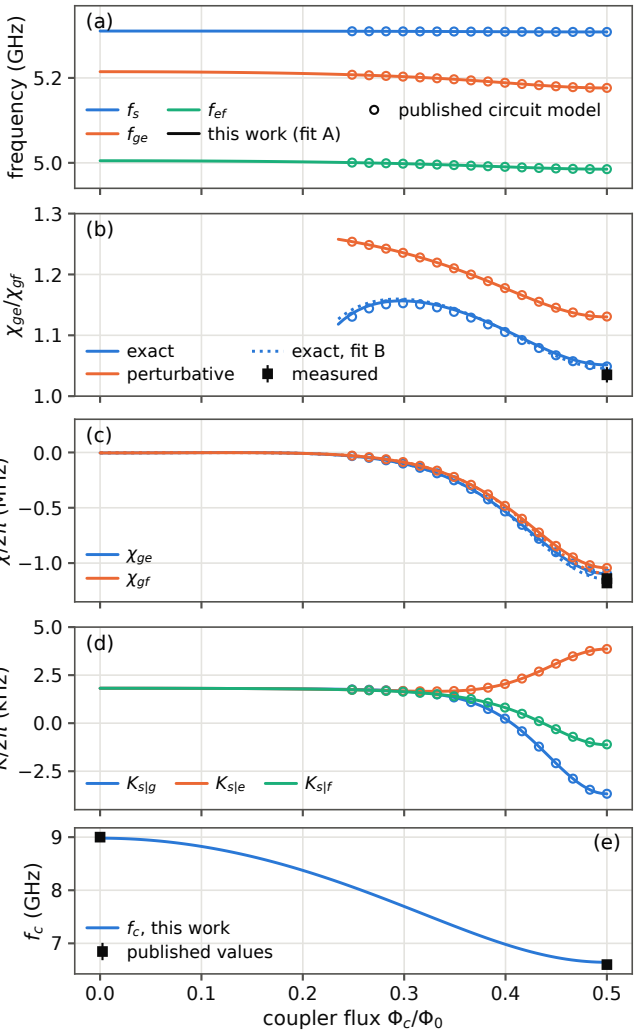
(a)
frequency (GHz)
5.2
5.0
f_s
f_{ef}
published circuit model
f_{ge}
this work (fit A)
(b)
χ_{ge}/χ_{gf}
1.3
1.2
1.1
1.0
exact
exact, fit B
perturbative
measured
(c)
$\chi/2\pi$ (MHz)
0.0
−0.5
−1.0
χ_{ge}
χ_{gf}
(d)
$K/2\pi$ (kHz)
5.0
2.5
0.0
−2.5
$K_{s|g}$
$K_{s|e}$
$K_{s|f}$
(e)
f_c (GHz)
9
8
7
f_c, this work
published values
0.0
0.1
0.2
0.3
0.4
0.5
coupler flux Φ_c/Φ_0